
Using the source provided, respond to all parts of the question.

1. Your response to the question should be provided in six parts: A, B, C, D, E, and F. Write the response to each part of the question in complete sentences. Use appropriate psychological terminology in your response.
 - A. Identify the research method used in the study.
 - B. State the operational definition of person-oriented dog behaviors.
 - C. Describe what the mean of the person-oriented behaviors indicates for the laughing trials as compared to the talking trials.
 - D. Identify at least one ethical guideline applied by the researchers.
 - E. Explain the extent to which the research findings may or may not be generalizable using specific and relevant evidence from the study.
 - F. Explain how at least one of the research findings supports or refutes the idea that dogs' expressions of the person-oriented behaviors demonstrate stimulus discrimination in operant conditioning.

Introduction

Dogs often comfort their owners by making visual and/or physical contact when the owners cry or by providing help to their owners when they are sick. The study examined whether a dog's reactions to a person's emotions differ based on the dog's prior experience with the person (owner versus stranger).

Participants

Researchers recruited community members with nonaggressive dogs via email. Of those recruited, 16 dog and owner pairs participated in this study. The owners ranged in age from 25 to 60 years (mean age = 47.06 years; standard deviation = 11.89 years), and the dogs ranged in age from 2 years to 13 years (mean age = 6.87 years; standard deviation = 3.35 years). The length of time the owners had their dogs before the study began ranged from 2 months to 13 years (mean = 5.70 years; standard deviation = 3.69 years). The study included the following dog breeds, with the number of each in parentheses: Lab mix (3), black Lab (1), terrier mix (2), corgi or corgi mix (3), French bulldog (1), Jack Russell terrier (1), miniature pinscher (1), miniature schnauzer (1), Shih Tzu (2), and wirehaired pointing griffon (1).

Method

Each dog owner received and signed a consent form. Researchers tested each dog at their owner's house, and the dogs received dog biscuits as compensation for participation in the study.

All dog and owner pairs participated in all four trials of the study. This is called a "withinsubjects" design, which means that researchers observe each participant in every condition of the study so that a participant can be directly compared to themselves across conditions. The "within-subjects" design serves the same purpose as random assignment.

A researcher who was unfamiliar with the dogs before the study played the role of the stranger. Upon entering each participant’s home, this “stranger-researcher” ignored the dogs by not interacting with them in any way. The stranger-researcher asked the owner to follow the same set of instructions during each trial:

“When you are asked to cry, please pretend to cry to the best of your ability for 20 seconds. The researcher will tell you when to begin and when to stop. When you are asked to laugh, please pretend to laugh to the best of your ability for 20 seconds. When you are acting, please use approximately the same volume. Also, please do not refer to your dog by name, look directly at them, or initiate or reciprocate physical contact during the study.”

The owners and the stranger-researcher stayed seated and moved their bodies naturally while they acted out the different emotional states, but they stayed seated. Each dog participated in four separate 20-second-long trials in which (1) the owner cried, (2) the stranger-researcher cried, (3) the owner laughed, and (4) the stranger-researcher laughed.

The order of these trials was counterbalanced across dogs, meaning that each dog completed the four trials in a randomized order. Before the trials began (baseline), in between each trial, and at the end of the fourth trial, the stranger-researcher and the owner had a light-hearted conversation for 2 minutes to reset the emotional experience for the dogs and the owners.

Results and Discussion

Researchers focused on two different kinds of dog behaviors. Person-oriented dog behaviors included looking at a person (either the owner or the stranger-researcher), making contact with a person (touching the person in some way), approaching a person, and vocalizing at a person (barking, whining, etc.). Non-person-oriented dog behaviors included passive behavior (e.g., lying down or sitting), walking, solitary play, and general vocalizing (not directed at a person). The total number of person-oriented behaviors from the dogs for each trial behavior the owners and researcher-strangers acted out is presented in the table.

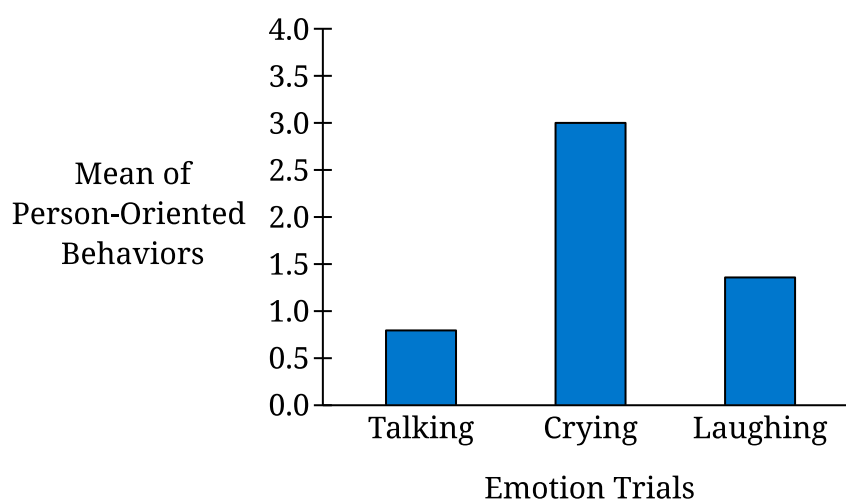
Total Number of Person-Oriented Dog Behaviors for Each Trial

Behavior	Dogs Looking	Dogs Making Contact	Dogs Approaching	Dogs Vocalizing	Total Person-Oriented Behaviors
Baseline	5	3	2	2	12
Owner crying	20	15	6	1	42
Owner laughing	11	2	2	0	15
Stranger crying	24	11	8	3	46

Behavior	Dogs Looking	Dogs Making Contact	Dogs Approaching	Dogs Vocalizing	Total Person-Oriented Behaviors
Stranger laughing	11	7	2	0	20
Totals	71	38	20	6	135

The graph displays the means for the number of person-oriented behaviors for the three types of emotions. The results are statistically significant ($p < 0.0011^1$):

Mean of Person-Oriented Behaviors for Emotion Trials



The results show that the crying trial elicited significantly more person-oriented behaviors than the laughing and talking trials. The mean of person-oriented behaviors when the owner and stranger-researcher were crying significantly differed from the means for laughing and talking, but the means for laughing and talking did not significantly differ from each other. Importantly, dogs primarily engaged with the individual who was crying, regardless of whether they were the owner or the stranger-researcher. In the owner-crying trial, dogs demonstrated person-oriented behaviors 75% of the time, and in the stranger-researcher-crying trial, dogs demonstrated the person-oriented behaviors 73% of the time.

Meyers-Manor, J. E., & Botten, M. L. (2020). A shoulder to cry on: Heart rate variability and empathetic behavioral responses to crying and laughing in dogs. *Canadian Journal of Experimental Psychology / Revue canadienne de psychologie expérimentale*, 74(3), 235–243.

1: p values indicate statistical significance. A p value of less than 0.05 is generally considered significant and increases the likelihood that the difference in means is due to the procedures of the study.